Telemarketing Campaign Predictive Analysis

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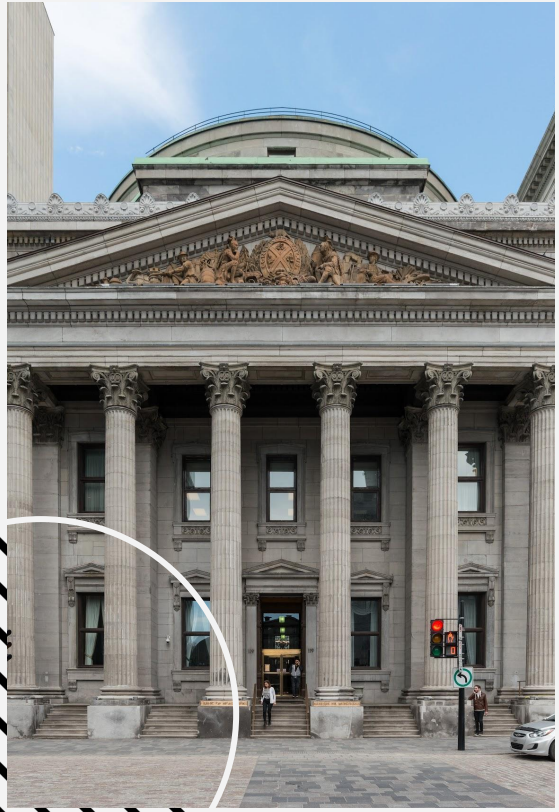

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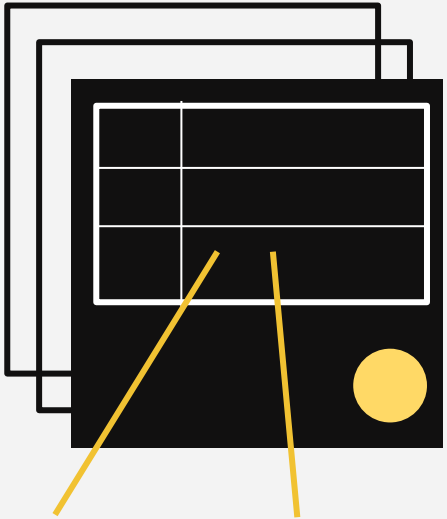
Introduction

1.1 Background

- Portuguese bank telemarketing campaign
- Promote financial product to clients
- Term deposit: fixed-term investment
- Predict whether clients will subscribe to term deposit

1.2 The Dataset



45,211

Entries (clients)

16

Features or independent variables

Subscribe to term deposit (YES/NO)

Target or response variable

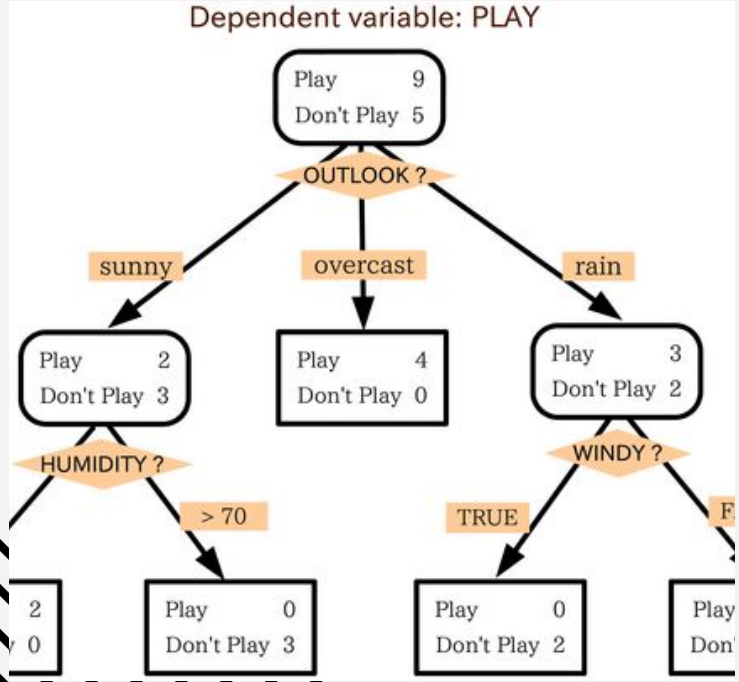
1.3 Problems

Classification problem: predict whether a client will subscribe to a term deposit based on the provided features

What are the **important factors** that influence a potential client's decision to subscribe to a term deposit?

How accurate are the predictive models in forecasting subscription outcomes?

How much do decision trees benefit from **bagging** (random forest) and **boosting** (XGBoost)?



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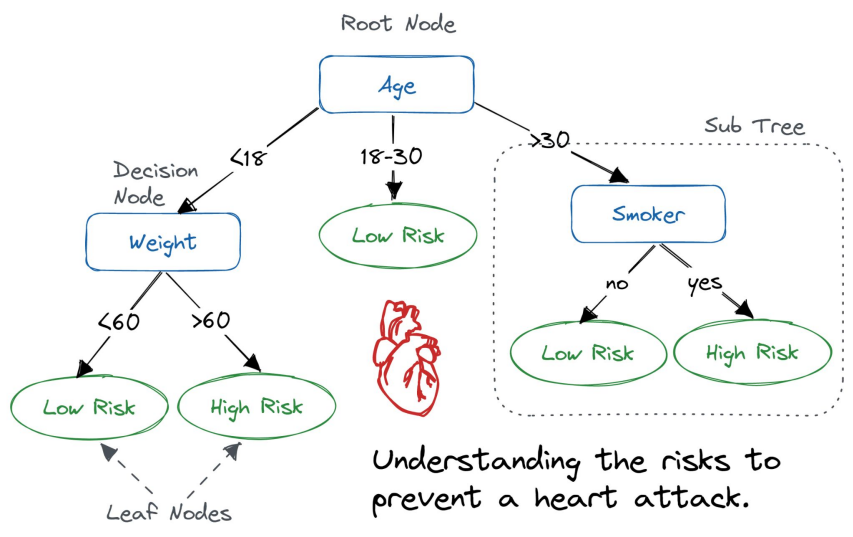
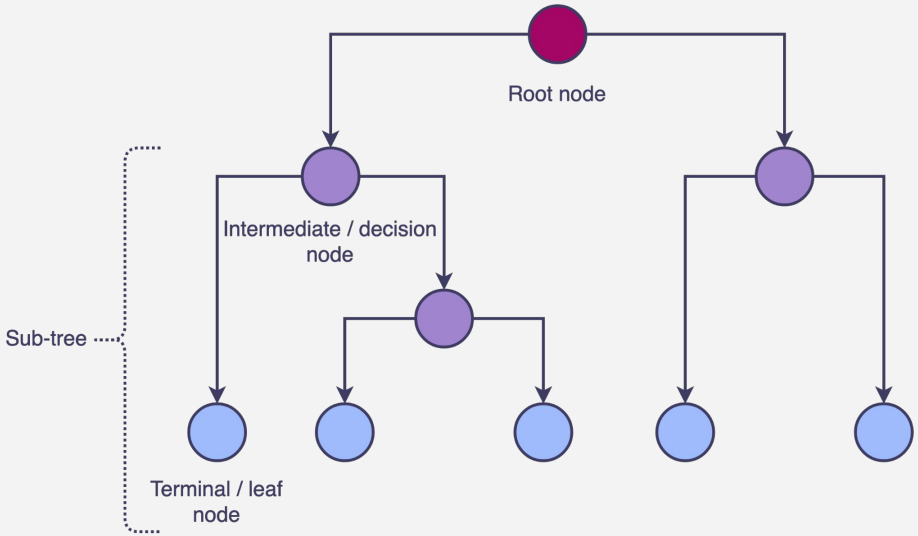
Tree-Based Models



- Predict by splitting predictor space
- <u>Prediction= leaf node with most samples</u>
- Advantages:
  - Simple, straightforward, interpretable
  - Mirrors human decision making
- Decision Tree, Random Forest, extreme gradient boosting (XGBoost)

2.1 Decision Tree



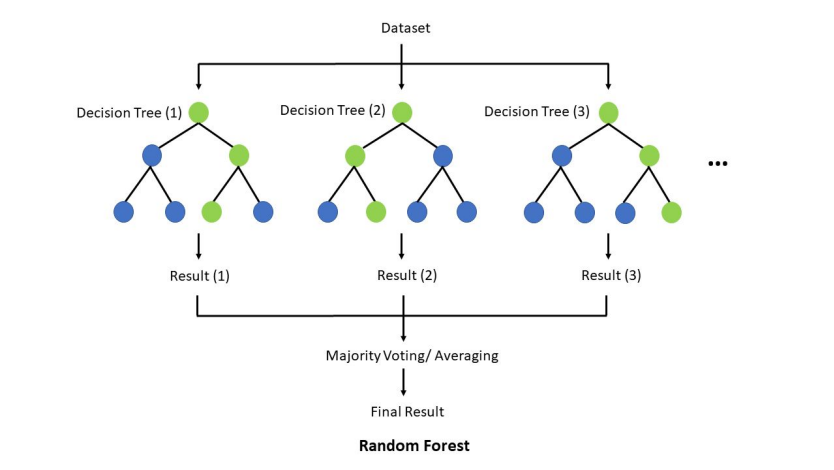
- Recursive binary splitting

$$T(X_i;\ \theta)$$

Predictor Threshold

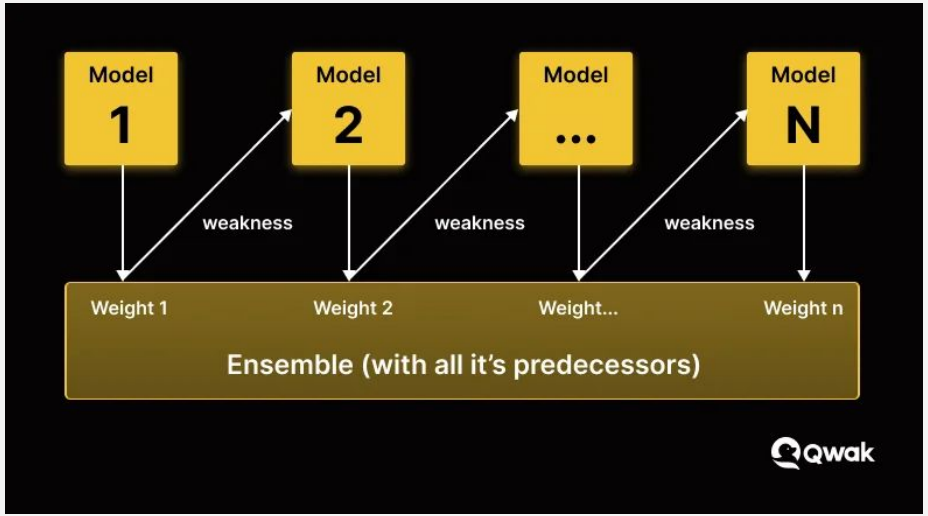
- Gini / entropy / information gain
- Baseline

2.2 Random Forest



- Construct multiple trees on different subsets of the data and features
- Final prediction is based on majority voting
- Diverse trees, reduce overfitting

2.3 XGBoost



- combines weak learners into a single strong learner by reducing a loss function iteratively.

$$F_m(x_i) = F_{m-1}(x_i) + \gamma_m h_m(x_i)$$

mth Model

m-1th Model

new tree

- Regularization term can be added to the Loss function to reduce overfitting

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Problem Solution

3.1 Data Preparation

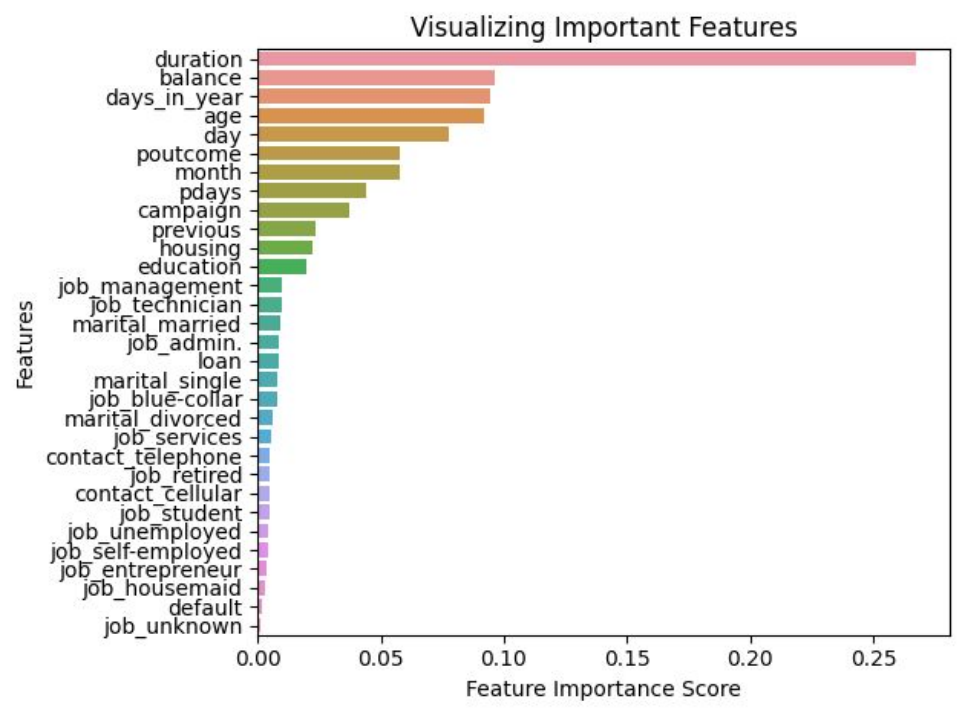
- Data Preprocessing
- Anomaly (Outlier) detection
- Exploratory Data Analysis (EDA)
  - Univariate
  - Multivariate analysis
- Feature Engineering: day_in_year

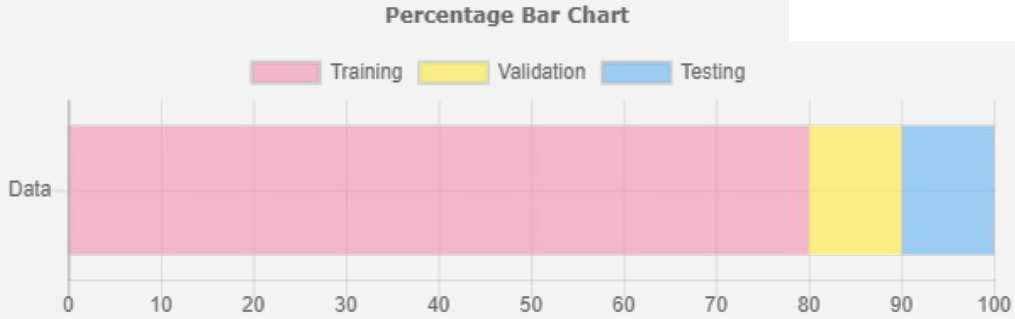
Month	Day	→	day_in_year
2	28	31+28 total days	**59**

More Preparation!

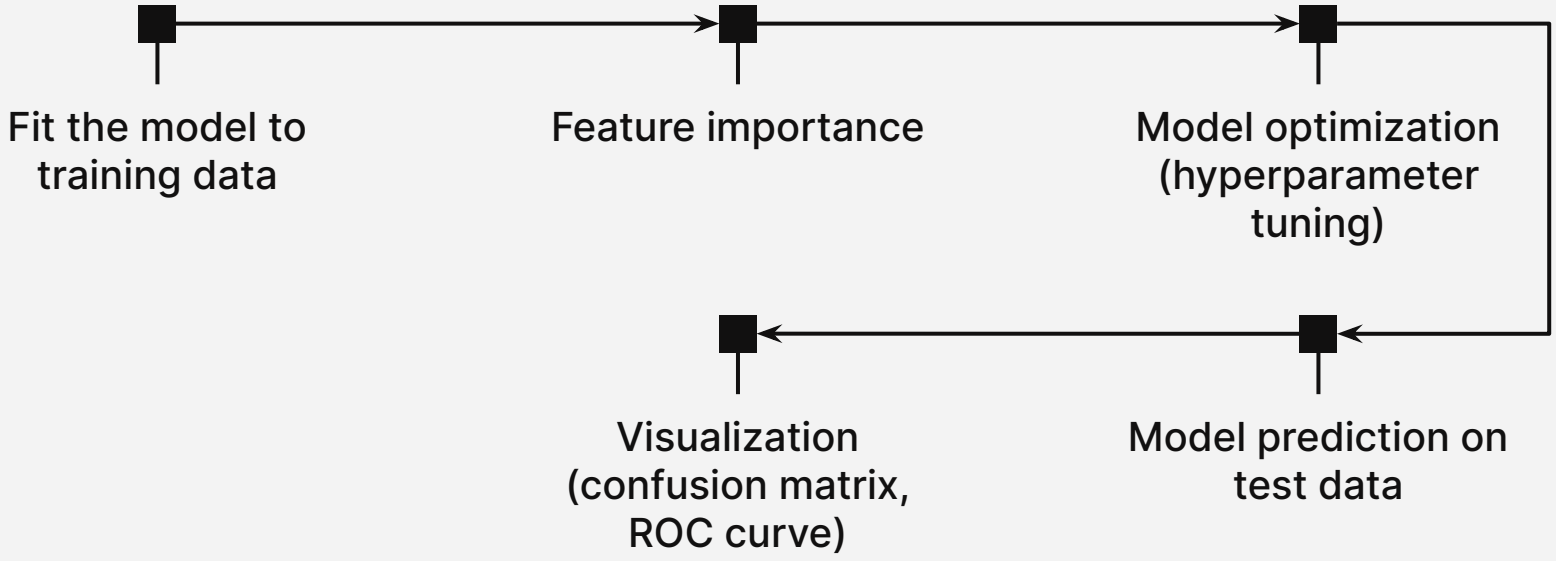
- Feature Importance

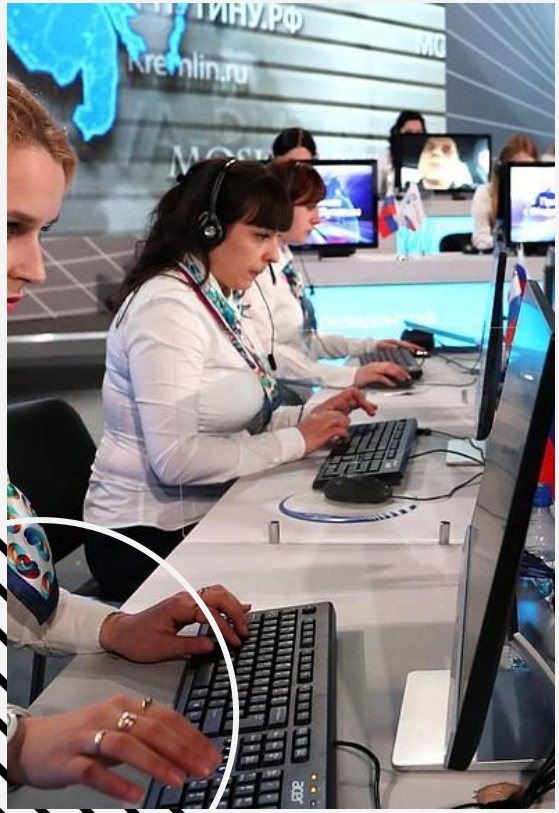
- Train-validation-test split (80% - 10% - 10%)





3.2 Machine Learning Modeling





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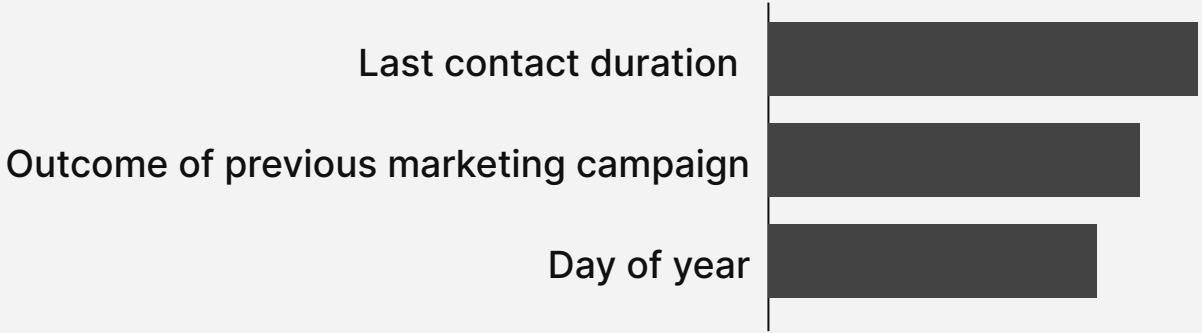
Result Interpretation

4.1 Key Insights

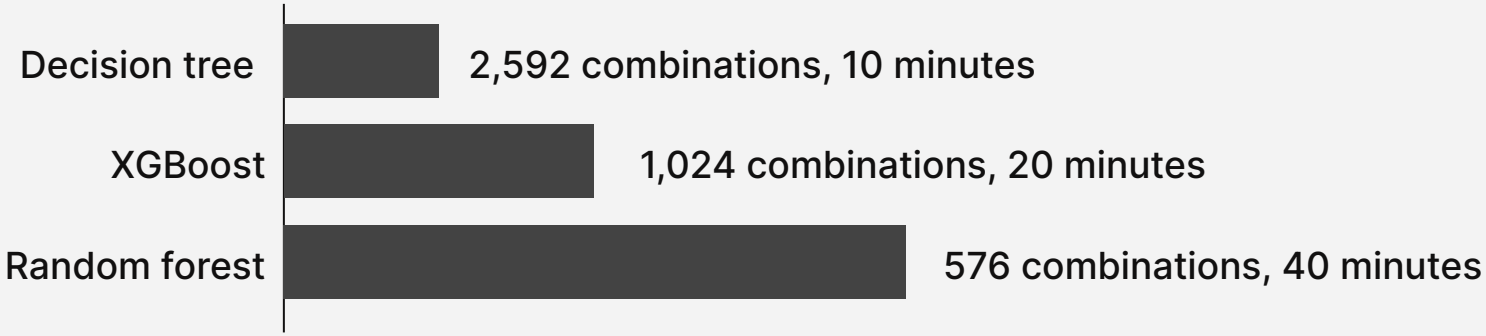
- Feature selection



- Training and testing accuracies

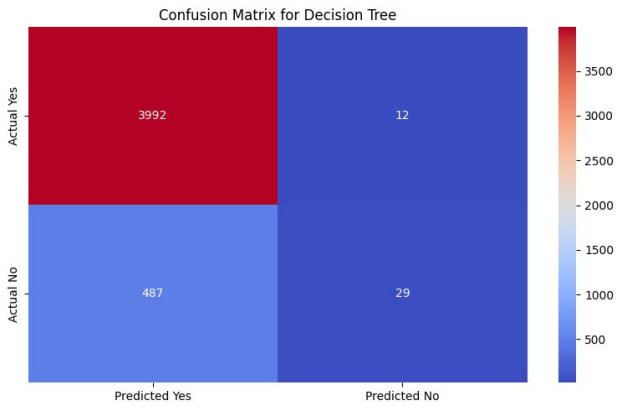
	Decision tree	Random forest	XGBoost
Training	89.95%	91.88%	92.47%
Testing	89.98%	90.44%	91.17%

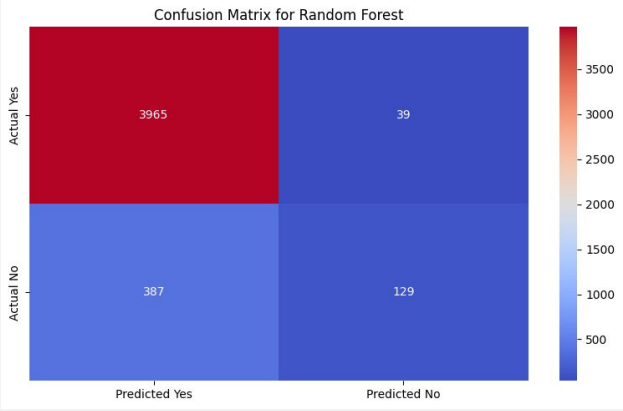
- Hyperparameter tuning computation time

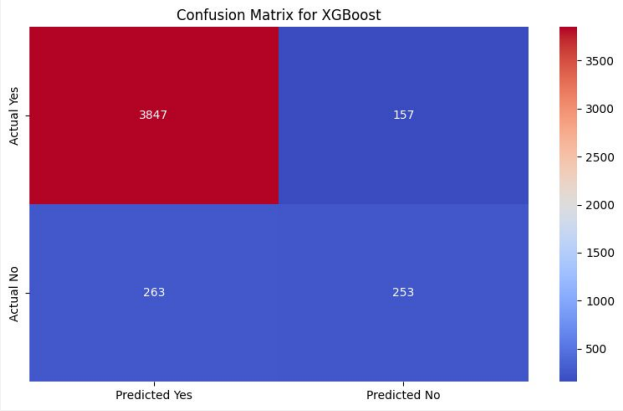


4.2 More Visualizations

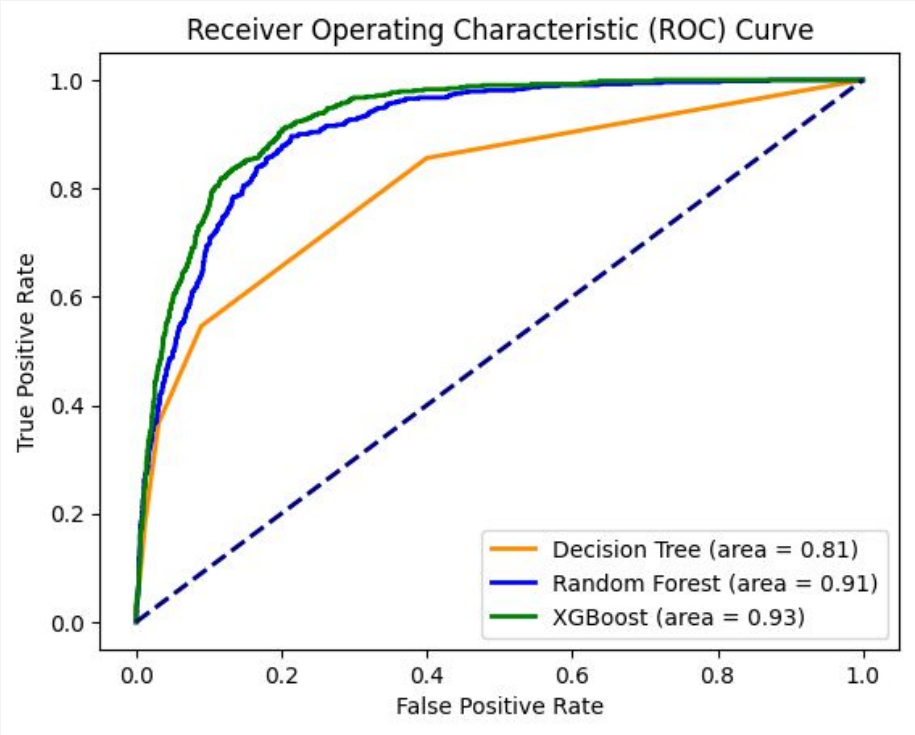
- Confusion matrix
  - Similar - most outcomes correctly predicted as positive ('Yes')
  - Compared to decision tree, random forest and XGBoost show more true negatives and false negatives but fewer false positives



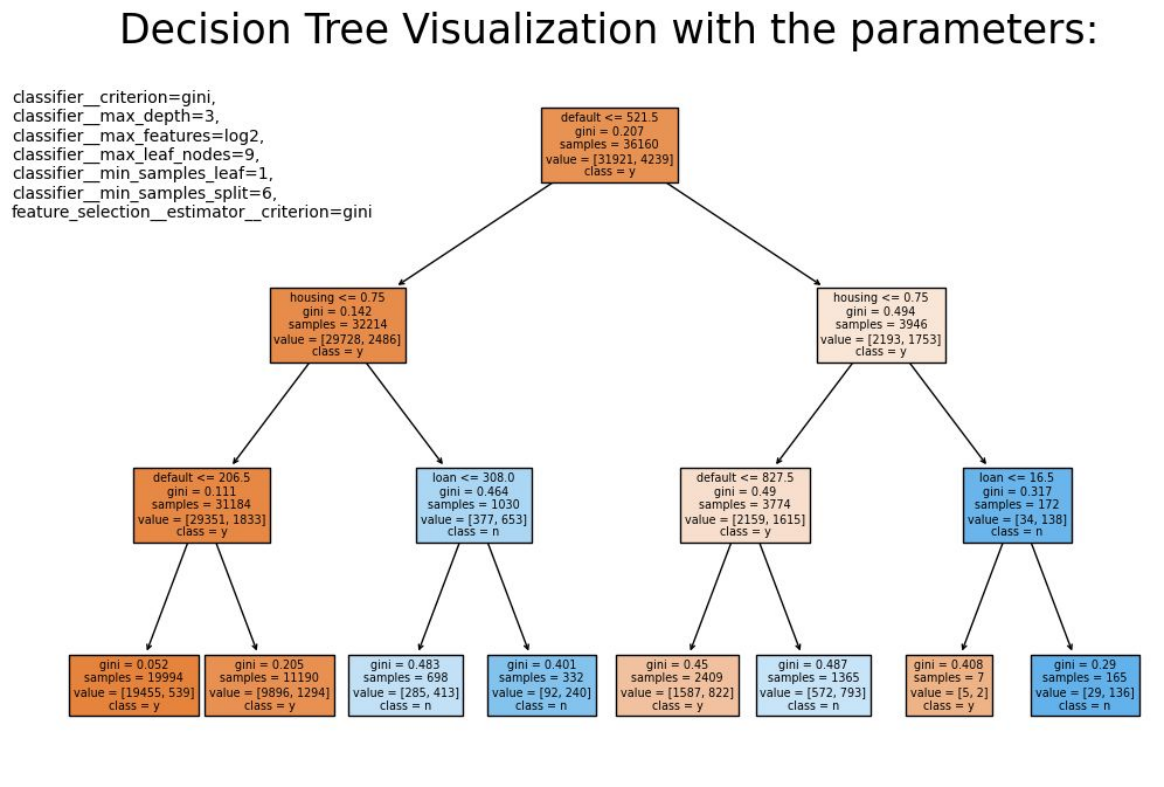




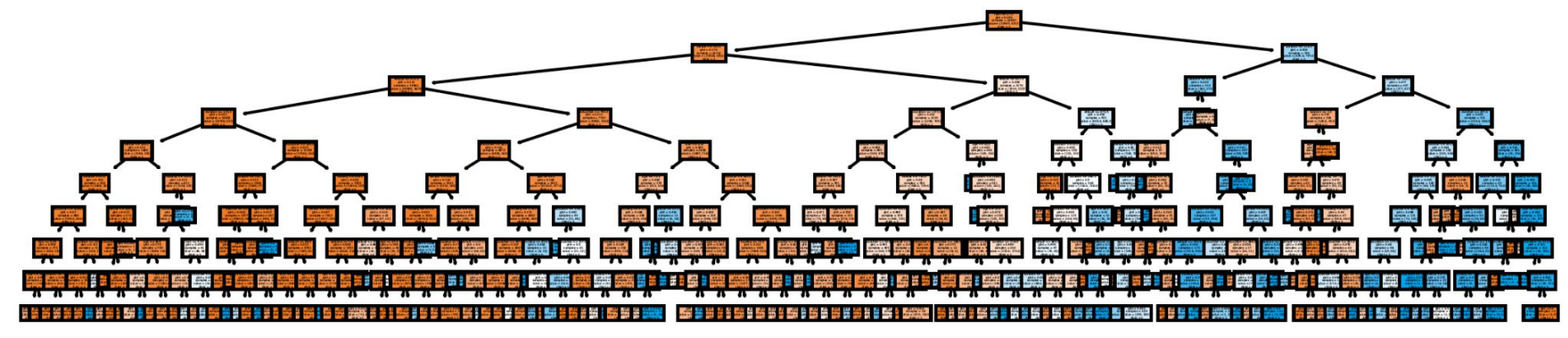
- Area under the ROC curve
  - XGBoost and random forest have larger AUC than decision tree
  - XGBoost and random forest are better at distinguishing between positive and negative classes

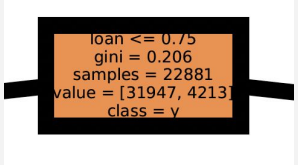


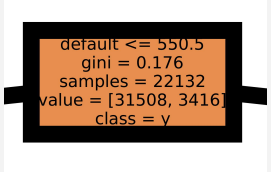
- Model visualization of Decision Tree

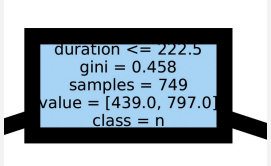


- Model visualization of Random Forest

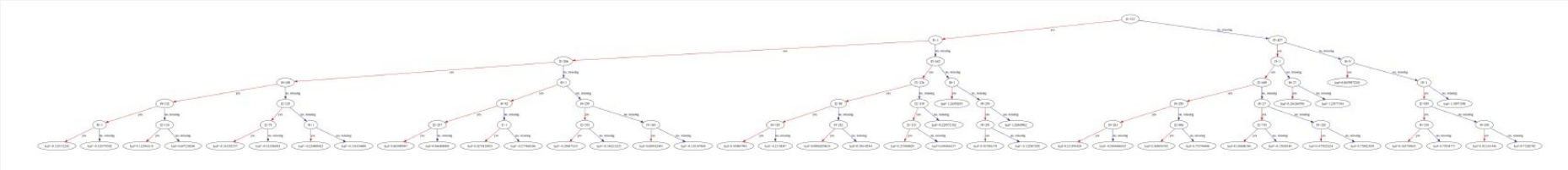


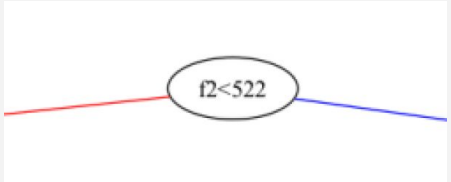




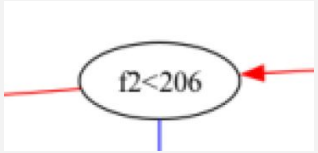


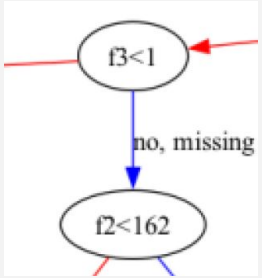
- Model visualization of XGBoost

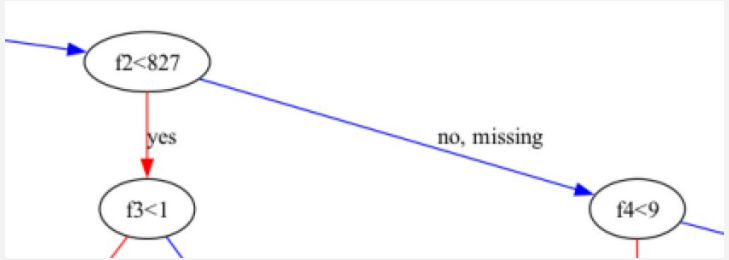




f2: duration
f3: campaign
F4: pdays







05

Conclusion

- Decision tree, random forest, and XGBoost models helped reveal **key predictors** for term deposit subscriptions

- Test accuracy scores ranging from **89% to 91%**, suggesting effective prediction capability

- Although effective, **ensemble methods** can make **interpretation** of individual predictions **challenging**

- If given more time - more hyperparameter tuning, find additional data, explore temporal patterns or demographic clustering

Thank You!